

Step-by-Step PostgreSQL Business Problem Solving Report

This report explains the PostgreSQL business problems solved using SQL queries after cleaning and loading the HR dataset into the hr_data table.

Step 1: View HR Dataset

```
SELECT * FROM hr_data;
```

Business Insight: Displayed all employee records stored in PostgreSQL.

Step 2: Find Null Values

```
SELECT *  
FROM hr_data  
WHERE employee_id IS NULL  
OR age IS NULL  
OR gender IS NULL  
OR department IS NULL;
```

Business Insight: Identified incomplete records before analysis.

Q1: Calculate Overall Attrition Rate

```
SELECT  
ROUND(  
SUM(CASE WHEN attrition='Yes' THEN 1 ELSE 0 END)  
*100/COUNT(*),2  
) AS Attrition_Rate  
FROM hr_data;
```

Business Insight: Measured overall employee attrition percentage.

Q2: Attrition Rate by Department

```
SELECT department,  
ROUND(  
SUM(CASE WHEN attrition='Yes' THEN 1 ELSE 0 END)  
*100/COUNT(*),2  
) AS Attrition_Rate  
FROM hr_data  
GROUP BY department;
```

Business Insight: Identified departments with highest attrition.

Q3: Average Salary by Job Role

```
SELECT job_role,  
ROUND(AVG(monthly_income::numeric),2) AS Average_Salary  
FROM hr_data  
GROUP BY job_role  
ORDER BY Average_Salary DESC;
```

Business Insight: Compared salary distribution across job roles.

Q4: Top 10 Highest Paid Employees

```
SELECT *
FROM hr_data
ORDER BY monthly_income DESC
LIMIT 10;
```

Business Insight: Found top-paid employees in the organization.

Q5: Rank Employees by Salary

```
SELECT employee_id,
monthly_income,
RANK() OVER(ORDER BY monthly_income DESC) AS salary_rk
FROM hr_data;
```

Business Insight: Ranked employees according to salary.

Q6: Employees Earning Above Average Salary

```
WITH avg_sal_cte AS(
SELECT AVG(monthly_income) AS avg_sal
FROM hr_data
)
```

```
SELECT *
FROM hr_data
WHERE monthly_income >
(
SELECT avg_sal
FROM avg_sal_cte
);
```

Business Insight: Identified employees earning above company average salary.

Q7: Identify Employee Churn Risk

```
SELECT employee_id,
department,
job_role,

CASE
WHEN job_satisfaction <=2
AND overtime='Yes'
AND work_life_balance <=2
AND performance_rating <=2
THEN 'High Churn Risk'

WHEN job_satisfaction =3
OR work_life_balance =3
THEN 'Medium Churn Risk'

ELSE 'Low Churn Risk'
END AS churn_risk
FROM hr_data;
```

Business Insight: Predicted employees likely to leave the company.

Q8: High Salary but Low Performance Employees

```
SELECT employee_id,  
department,  
monthly_income,  
performance_rating  
FROM hr_data  
WHERE monthly_income >  
(  
SELECT AVG(monthly_income)  
FROM hr_data  
)  
AND performance_rating <=2;
```

Business Insight: Detected highly paid but low-performing employees.

Q9: Does Overtime Increase Attrition?

```
SELECT overtime,  
COUNT(*) AS total_employees,  
SUM(  
CASE WHEN attrition='Yes'  
THEN 1 ELSE 0  
END  
) AS attrition_count,  
ROUND(  
SUM(CASE WHEN attrition='Yes'  
THEN 1 ELSE 0 END)  
*100.0/COUNT(*),2  
) AS attrition_rate  
FROM hr_data  
GROUP BY overtime;
```

Business Insight: Analyzed overtime impact on attrition.

Q10: Job Roles with Lowest Satisfaction

```
SELECT job_role,  
ROUND(AVG(job_satisfaction::numeric),2) AS avg_satisfaction  
FROM hr_data  
GROUP BY job_role  
ORDER BY avg_satisfaction ASC;
```

Business Insight: Detected job roles with low employee satisfaction.

Q11: Are High Salaries Reducing Attrition?

```
SELECT  
CASE  
WHEN monthly_income < 30000 THEN 'Low Salary'  
WHEN monthly_income BETWEEN 30000 AND 70000 THEN 'Medium Salary'  
ELSE 'High Salary'  
END AS salary_category,  
COUNT(*) AS total_employees,  
SUM(  
CASE WHEN attrition='Yes'  
THEN 1 ELSE 0  
END
```

```

) AS attrition_count,
ROUND(
SUM(
CASE WHEN attrition='Yes'
THEN 1 ELSE 0
END
)*100.0/COUNT(*),2
) AS attrition_rate
FROM hr_data
GROUP BY salary_category;

```

Business Insight: Analyzed salary impact on employee retention.

Q12: Which Age Group Leaves Most?

```

SELECT
CASE
WHEN age BETWEEN 18 AND 25 THEN '18-25'
WHEN age BETWEEN 26 AND 35 THEN '26-35'
WHEN age BETWEEN 36 AND 45 THEN '36-45'
ELSE '46+'
END AS age_group,
COUNT(*) AS attrition_count
FROM hr_data
WHERE attrition='Yes'
GROUP BY age_group
ORDER BY attrition_count DESC;

```

Business Insight: Identified age groups with highest attrition.

Q13: Identify High-Risk Employees

```

SELECT employee_id,
department,
job_role,
monthly_income,
CASE
WHEN job_satisfaction <=2
AND performance_rating <=2
AND work_life_balance <=2
AND overtime='Yes'
THEN 'High Risk'
ELSE 'Normal'
END AS employee_risk

FROM hr_data;

```

Business Insight: Identified high-risk employees likely to leave.

Conclusion

Using PostgreSQL, important HR business problems were solved through SQL analysis. The queries helped identify employee attrition trends, salary analysis, employee risk detection, overtime impact, and workforce insights. These insights were later used in Power BI dashboards.